VANISHING OF THE FIRST COHOMOLOGY AND THE RECOVERY OF AN INVERTIBLE MODULE FROM ITS DIRECT IMAGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 9429-9451 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 9453-9516, the complete author proof (62 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Flat proper families with f_*O_X = O_S, cohomology and base change, and the norm or trace of an invertible module; the argument is retained.

Source reference. Original lemma lemma-diagonal-picard-flat-proper: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9199–9215. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma more-algebra-lemma-lift-perfect-from-residue-field: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9516–9516. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma perfect-lemma-affine-compare-bounded: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9516–9516. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma perfect-lemma-flat-proper-perfect-direct-image-general: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9516–9516. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma perfect-lemma-perfect-affine: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9516–9516. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Consider a commutative diagram of schemes*

$$\begin{array}{ccc} X' & \longrightarrow & X \\ & {\scriptstyle f'} \searrow \quad \swarrow {\scriptstyle f} & \\ & S & \end{array}$$

with $f' : X' \to S$ and $f : X \to S$ satisfying the hypotheses of Lemma lemma-diagonal-picard-flat-proper. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module and let $\mathcal{L}'$ be the pullback to X'. Let $Z \subset S$, resp. $Z' \subset S$ be the locally closed subscheme constructed in Lemma lemma-diagonal-picard-flat-proper for $(f, \mathcal{L})$, resp. $(f', \mathcal{L}')$ so that $Z \subset Z'$. If $s \in Z$ and

$$H^1(X_s, \mathcal{O}) \longrightarrow H^1(X'_s, \mathcal{O})$$

is injective, then $Z \cap U = Z' \cap U$ for some open neighbourhood U of s.

Proof. We may replace S by Z'. After shrinking S to an affine open neighbourhood of s we may assume that $\mathcal{L}' = \mathcal{O}_{X'}$. Let $E = Rf_*\mathcal{L}$ and $E' = Rf'_*\mathcal{L}' = Rf'_*\mathcal{O}_{X'}$. These are perfect complexes whose formation commutes with arbitrary change of base (Derived Categories of Schemes, Lemma perfect-lemma-flat-proper-perfect-direct-image-general). In particular we see that

$$E \otimes^{\mathbf{L}}_{\mathcal{O}_S} \kappa(s) = R\Gamma(X_s, \mathcal{L}_s) = R\Gamma(X_s, \mathcal{O}_{X_s})$$

The second equality because $s \in Z$. Set $h_i = \dim_{\kappa(s)} H^i(X_s, \mathcal{O}_{X_s})$. After shrinking S we can represent E by a complex

$$\mathcal{O}_S \to \mathcal{O}_S^{\oplus h_1} \to \mathcal{O}_S^{\oplus h_2} \to \ldots$$

see More on Algebra, Lemma more-algebra-lemma-lift-perfect-from-residue-field (strictly speaking this also uses Derived Categories of Schemes, Lemmas perfect-lemma-affine-compare-bounded and perfect-lemma-perfect-affine). Similarly, we may assume E' is represented by a complex

$$\mathcal{O}_S \to \mathcal{O}_S^{\oplus h'_1} \to \mathcal{O}_S^{\oplus h'_2} \to \ldots$$

where $h'_i = \dim_{\kappa(s)} H^i(X'_s, \mathcal{O}_{X'_s})$. By functoriality of cohomology we have a map

$$E \longrightarrow E'$$

in $D(\mathcal{O}_S)$ whose formation commutes with change of base. Since the complex representing E is a finite complex of finite free modules and since S is affine, we can choose a map of complexes

$$\begin{array}{ccccccc} \mathcal{O}_S & \xrightarrow{d} & \mathcal{O}_S^{\oplus h_1} & \longrightarrow & \mathcal{O}_S^{\oplus h_2} & \longrightarrow & \ldots \\ \downarrow{\scriptstyle a} & & \downarrow{\scriptstyle b} & & \downarrow{\scriptstyle c} & & \\ \mathcal{O}_S & \xrightarrow{d'} & \mathcal{O}_S^{\oplus h'_1} & \longrightarrow & \mathcal{O}_S^{\oplus h'_2} & \longrightarrow & \ldots \end{array}$$

representing the given map $E \to E'$. Since $s \in Z$ we see that the trivializing section of $\mathcal{L}_s$ pulls back to a trivializing section of $\mathcal{L}'_s = \mathcal{O}_{X'_s}$. Thus $a \otimes \kappa(s)$ is an isomorphism, hence after shrinking S we see that a is an isomorphism. Finally, we use the hypothesis that $H^1(X_s, \mathcal{O}) \to H^1(X'_s, \mathcal{O})$ is injective, to see that there exists a $h_1 \times h_1$ minor of the matrix defining b which maps to a nonzero element in $\kappa(s)$. Hence after shrinking S we may assume that b is injective. However, since $\mathcal{L}' = \mathcal{O}_{X'}$ we see that $d' = 0$. It follows that $d = 0$. In this way we see that the trivializing section of $\mathcal{L}_s$ lifts to a section of $\mathcal{L}$ over X. A straightforward topological argument (omitted) shows that this means that $\mathcal{L}$ is trivial after possibly shrinking S a bit further. □